# Writing a note

NSDF-Data student guide 3 — writing a note, for every computer

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**Tip**

**This is guide 3 of 5.** It is about *writing* the note. Guide 4 covers *publishing* it, and guide 5 covers publishing with an AI assistant.

**Tip**

**What a note is.** A note is a short, honest write-up of one question you investigated: what you asked, what you did, what you found, and how sure you are. Notes are how the lab keeps a record that someone else can follow and check. You do not need to be an expert to write one. You do need to be specific.

## Before you start

- You have finished the setup guide and the first analysis for your computer.
- You know which question you are answering. If you do not, pick a small one from the project's *Next steps* sections (for example, a number you can reproduce from a note).
- You have a notebook (`.ipynb`) or a script that produces the numbers and figures you will show. Keep it. The note points to it.

## What a note is for

A reader should be able to answer four questions after reading your note:

1. **What was the question?**
2. **What data and code did you use?**
3. **What did you find, and how sure are you?**
4. **What would you do next?**

If your note does not answer these, it is not finished, however long it is.

## The parts of a note

**Only the header is required.** Every other section is a suggestion. Use the sections that help your reader, and leave out the ones that do not apply. A short note with a header and one clear result is complete. A long note with no caveats is not.

The table below lists the sections most notes use, in a sensible order. Each one you use becomes a heading, with an outline link.

| Section | What goes in it | Length |
|---|---|---|
| **Title** | The question, as a sentence. “Why does X happen in Y?” is better than “Analysis of X.” | one line |
| **Header** | Author, date, status, and one-sentence description (see below). | a few lines |
| **Code repository** | The notebook and library files you used, each pinned to the commit that produced the results. | a short list |
| **Data and setup** | Which series, dump and detector; which quantities; which windows and cuts. Define any word a newcomer would not know. | one or two paragraphs |
| **The question** | The question again, and why it matters. | a paragraph |
| **Results** | Your figures and tables, each with a caption that says what the reader is looking at. | as needed |
| **Reading** | What you think the results mean, and what they do *not* show. | a few bullets |
| **Caveats** | What could be wrong. Small sample? One channel only? A cut you chose by eye? | a short list |
| **Next steps** | Concrete things to do next. | a short list |

**Tip**

**Caveats are strongly suggested.** A note that states its limits is trusted more than one that does not, however good the plots look. If you leave caveats out, be sure there is nothing you would want a reader to know about your result.

## Where and how to write

Write your note in whatever you are comfortable with. Two routes are supported, and guide 4 takes either one onto the site.

### Suggested: VS Code, writing Markdown in the project

VS Code is a free editor. Here you write the note as a **plain text file ending in** `.md` inside the `NSDF-Data` folder, next to the notebook that makes its numbers. It can show a preview of the page as you type, and everything stays in Git on your branch, so every version is saved and changing the note

later is just editing the file. This is the route we suggest.

1. Install VS Code, start it, and choose **File**, then **Open Folder**, and pick the `NSDF-Data` folder.
2. In the file list on the left, open `notes/src`. Copy `_template.md` (right-click it, **Copy**, then right-click the folder and **Paste**) and rename the copy `your-note-name.md`. Use lowercase letters and hyphens, for example `baseline-in-dump-one.md`.
3. Write. To see the preview in its own tab next to the text, press `Ctrl+K`, let go, then press `V` (on macOS: `Cmd+K`, then `V`). The preview follows as you type and scroll. (`Ctrl+Shift+V` also works, but it replaces the text with the preview in the same tab.)
4. Save each figure as a PNG in `notes/src/img/`, with your note's name at the start of the file name (`baseline-in-dump-one-fig1.png`) and refer to it as `![caption](img/baseline-in-dump-one-fig1.png)`. All notes share that one folder, so the name prefix keeps your pictures from being mixed up with someone else's. Put the picture directly in `img/`, not in a subfolder.
5. Keep your notebook and any scripts in your own folder under `data_analysis/R76/student/`, and commit as you go (guide 2, Step 7).

Any other plain text editor works for this route too (Notepad, TextEdit set to **Format**, then **Make Plain Text**, or gedit). Save the file as `your-note-name.md`, as UTF-8, and check on Windows that the name does not end in `.md.txt` (File Explorer, **View**, then tick **File name extensions**). The GitHub web editor also works, with no install at all.

#### Also supported: Word or Google Docs

If you write better in Word or Google Docs, write there. Keep the file anywhere outside the project, for example your `Research` folder. When it is ready, guide 4 turns it into a note, pictures included, with one command, and the same command updates the note after you change the document. Three habits make the conversion clean:

- **Use Heading styles for section titles.** In Word, select a section title and choose **Home**, then **Heading 1** (main sections) or **Heading 2** (sub-sections). Do not just make the title bold or bigger. Only Heading-styled text becomes a heading, and without headings there is no outline.
- **Avoid underline and superscript.** Underlined text and superscripts (the raised "th" in 5th, for example) do not convert. Use plain text instead.
- **Insert pictures into the document** with **Insert**, then **Pictures** (or paste them in), and put the caption in the line under each one. Pictures that are only linked, not embedded, are lost.

Keep code out of the document where you can: link to the notebook instead (see *Pinning your code*). Word turns straight quotes into curly ones, which breaks any code you paste in.

## The header

Start your note file with a short block like this. These are the fields the site will read:

```
---
id: "TBD"
title: "Why does the 1000-sample window see more pulses?"
author: "Your Name"
date: "2026-10-05"
status: "In progress"
keywords: "baseline, pretrigger window, dump 1"
description: "One sentence: what you did and what you found."
```

```
---
```

Leave `id` as `TBD`. The reviewer gives your note its number (S1, S2, …) before it is published.

**Status** is one of: *In progress* (you are still working on it), *Complete* (the question is answered and you are not going to change it), or *Superseded (reference/link)* (a later note has taken its place; this one stays up for reference and links to the newer note). New notes start as *In progress*.

## If your note replaces an earlier note

Sometimes a new note redoes an earlier one with a better method. Then the two notes have to point at each other, and the older note's status has to change. Do this:

1. **In your note**, directly under the header, add one line that names the note you replace and links to it. Pages sit in the same folder, so a link to a hand-written note looks like this (note 3 here), and a link to another student note is `student-their-note-name.html`:

   ```
   **Supersedes:** [Note 3](note-03-noise-psd.html)
   ```

2. **In the pull request description** (guide 4), write `Supersedes Note 3` so your lead sees it at once.

3. **The older note** then changes to the status *Superseded (reference/link)* and gets a link to yours. Your lead does this for a hand-written note, in the same pull request, as an extra commit. For another student note, change `status:` in its header in `notes/src/` yourself, add a line `**Superseded by:** [your note](student-your-note-name.html)` under its header, and include that file in your pull request.

Do not delete the older note. It stays up for reference.

## Writing the body

- **Say what you did in order,** with enough detail that someone could repeat it: the file, the detector, the window length, the cut.
- **Use the project's words.** A *segment* is a stretch of consecutive events with the same trigger type. A *run* is a whole data-taking period. Do not use *run* for a segment.
- **Define each quantity once,** in the *Data and setup* section, and use the same name every time after. If you call it `bstd` in one place and "baseline noise" in another, the reader will think they are different things.
- **Write short sentences.** Put the number first, then what it means: "The median is 1.53 counts, so a typical trace is flat noise."
- **Say how sure you are.** "The two groups look the same" is weaker than "the medians are 1.51 and 1.58, from 141 and 38 traces." Give the counts.
- **Separate what you measured from what you think.** A sentence in *Results* states a number. A sentence in *Reading* says what it might mean.

**Tip**

If you change your mind, say so. Notes that show a correction are more useful than notes that pretend nothing went wrong. Put the correction in a clearly labelled box, with the date (see *A coloured box*, next).

### A coloured box

For a correction, a caveat or anything a reader must not miss, put it in a coloured box. In a Markdown note, write the text between two lines that hold only three colons, with a blank line before and after:

```
::: callout
**Correction, 12 October 2026.** The median in the first table
used 1000 samples; with 500 it is 1.51 counts.
:::
```

Start the box with a bold label such as **Correction**, **Caveat** or **Note**. All boxes have the same colour (shaded, with a border); only the label tells the reader which kind it is.

**Careful**

**The VS Code preview does not draw the box.** It shows the two `:::` lines as plain text. That is expected: the box appears on the published page, not in the preview. Do not delete the `:::` lines.

If you write in Word or Google Docs, a coloured box does not survive the conversion. Use a bold label such as **Correction.** at the start of the paragraph, or add the `:::` lines to the Markdown after converting.

## Figures and tables

- **Every figure needs a caption** that says what is plotted, for which data, and what to look for. "Bstd against event number, dump 1. Grey: trigger type Unknown." is a good caption.
- **Label the axes with units.** ADC counts, microseconds, or samples. Do not leave the units out.
- **Save figures as PNG** from your notebook, at a readable size, into the note's image folder (`notes/src/img/`, named as in *Where and how to write*) or, in a Word document, into the document.
- **Use a table** when you are comparing numbers. Use a figure when you are showing shape.

## Pinning your code

A note is only useful if the reader can find the code that made its numbers. Link to your notebook and library files at a **specific commit**, not at a branch name such as `develop` or `master`, because branches keep changing:

1. Commit your notebook to your branch (see the first analysis guide).
2. Find the commit's short code, for example `a1b2c3d`.
3. Link to `https://github.com/villano-lab/NSDF-Data/blob/a1b2c3d/data_analysis/R76/`

Say in the note which commit you pinned.

## Outline and links

If your note has more than two or three sections, add a short outline near the top, with a link to each one. Readers use it to jump to what they need. The site builds the outline from your ## headings, so use one ## heading for each section. A note with no ## headings simply has no outline.

## Checking your note before you submit

Go through this list before you ask for review:

- ☐ The title is a question.
- ☐ Every number in the text can be found in your notebook output.
- ☐ Every figure has a caption with units and a sentence on what to look for.
- ☐ Every notebook link is pinned to a commit.
- ☐ The header is complete.
- ☐ Any section you use says something a reader needs. Empty or filler sections are removed.
- ☐ You have said what you did *not* check, if the note makes a claim.
- ☐ If you wrote next steps, they are concrete.
- ☐ You have read it aloud once, looking for a word a newcomer would not know.

## Session Info

Guide version 12, 12 October 2026. Written for every computer. Please tell the project lead where a step was unclear.

## Next

When the note is written and checked, go to **student guide 4: publishing a note**.

## Where to get help

**Anthony Villano**, project lead: anthony.villano@ucdenver.edu

When you write, include the note's title, the step or section you are stuck on, and the sentence or number that worries you. Nobody will be annoyed by a question about a note that is not finished.

## Key links

- Notes site: https://villano-lab.github.io/NSDF-Data/
- Repository: https://github.com/villano-lab/NSDF-Data
- Example of a good note with a correction box: https://villano-lab.github.io/NSDF-Data/notes/note-02a-bstd-vs-time.html